

Multiple Choice (10 marks)

1. Which of the following spreadsheet functions would be most useful for detecting im-probably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row
2. Let `l = [1,2,3,4]`. What is `max(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
3. What is the output of `4*1**4` in python?
 - (a) 4
 - (b) 1
 - (c) 256
 - (d) 16
4. In Python 3, what is the output of `print list[1:3]` if `list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]`?
`786 , 2.23, 'john', 70.2`
 - (a) `abcd`
 - (b) `786, 2.23`
 - (c) `None of the above.`
5. The boolean expression `a[i] == max || !(max != a[i])` can be simplified to
 - (a) `a[i] == max`
 - (b) `a[i] != max`
 - (c) `a[i] < max || a[i] > max`
 - (d) `FALSE`

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the function `random()` sets the integer starting value used in generating random numbers.
7. True or False: In Python, the expression `r(3)` evaluates to 6 when `r` is defined as a lambda function that doubles its input.
8. True or False: Authentication is compromised when someone uses a stolen login and password to access a system.
9. True or False: In Python 3, the output of `set([1,2,2,3,4])` is `{1,2,3,4}`, which removes duplicate values.
10. True or False: The condition `a < b` is sufficient to ensure that the expression evaluates to true.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the square root of a perfect number.
12. Write a function to find the median of two sorted arrays of same size.

End of Paper